EXCELLENCE

Class 11 Physics · Session 2026–27

Vectors

Complete formula notes — Lectures 01 to 05

Full coverage of the Vectors block: types, components, resultant, dot product, cross product, work and torque. Every formula is on its own line. Symbols are explained before use.

How to use each page
1. Read “In plain English”
2. Read “What the words mean”
3. Copy each formula line separately
4. Try the green example

Full chapter map

0. Symbol dictionary
1. Scalar vs vector and notation
2. Types of vectors
3. Magnitude, unit vector, scalar multiply
4. Rectangular components and compass angles
5. Triangle law and resultant
6. Subtraction, component add, Δv
7. Dot product, projection, angle tricks
8. Cross product and $\hat{i}\,\hat{j}\,\hat{k}$
9. Physics uses — work, torque, more
10. Revision lock lines

0. Symbol dictionary — read this first

Symbol / short name	Full meaning in school language
A → or bold A	a vector (size + direction)
|A| or A	magnitude = length of the arrow (a number ≥ 0)
Â	unit vector of A (size 1, same direction)
$\hat{i}$, $\hat{j}$, $\hat{k}$	unit vectors along +x, +y, +z
A_x, A_y, A_z	components of A along the axes
θ	angle between two vectors (usually 0° to 180°)
R = A + B	resultant = vector sum
α	angle that R makes with A (in triangle law)
A · B	dot (scalar) product — answer is a number
A × B	cross (vector) product — answer is a vector
kA	scalar multiply — scale size; reverse if k < 0
Δv	change in velocity = $v_f - v_i$ (a vector)
W , τ	work (scalar), torque (vector)

Remember: If a formula uses a letter you forgot, open this page first.

1. Scalar vs vector and notation

In plain English: a scalar has only size. A vector has size and direction, and it must obey the triangle law when you add it. Having a "direction word" is not enough — electric current has a sense of flow but is not a vector.

What the words mean
- Magnitude = how long the arrow is
- Direction = which way the arrow points
- Triangle law = head-to-tail joining when you add vectors
- Negative of A (−A) = same size, opposite direction

Vector = magnitude × unit vector in its direction
$$\mathbf{A} \rightarrow = |A|\,\hat{A}$$

Scalars (size only)

Mass, time, temperature, energy

Speed (how fast, no direction)

Work and power are scalars

Current is not a vector

Vectors (size + direction)

Displacement, velocity, acceleration

Force, momentum, torque

Electric field, magnetic field

Reverse direction → reverse the arrow

Remember: $\hat{i}$, $\hat{j}$, $\hat{k}$ have size 1. They only carry direction along the axes.

2. Types of vectors — say these in one line

Name	Meaning in plain words
Equal vectors	Same kind, same size, same direction
Parallel	Same direction (sizes may differ)
Anti-parallel	Opposite directions
Negative of A	Same size, opposite direction: −A
Unit vector	Size = 1, keeps the direction

Name	Meaning in plain words
Null (zero)	Size = 0; no useful direction
Collinear	On the same line (or parallel lines)
Coplanar	Lie in the same plane
Orthogonal	Perpendicular (angle = 90°)
Concurrent	Lines of action meet at one point
Axial	Along an axis (τ, ω, L) — sense by RHR

Angle between two vectors = smaller angle between directions (0° to 180°)
Put both arrows tail-to-tail (or tip-to-tip) before you measure θ

3. Magnitude, unit vector, scalar multiply

In plain English: magnitude is the length. Unit vector keeps only direction. Multiplying by a number k stretches or shrinks; negative k flips the arrow.

What the words mean
- |A| = length of A (never negative)
- Â has no physical unit — only direction
- k > 0 → same direction; k < 0 → opposite direction

$$|\mathbf{A}| = \sqrt{(A_x^2 + A_y^2 + A_z^2)}$$
$$\hat{\mathbf{A}} = \mathbf{A} / |\mathbf{A}|$$
$$\hat{\mathbf{A}} = (A_x/|\mathbf{A}|)\,\hat{\imath} + (A_y/|\mathbf{A}|)\,\hat{\jmath} + (A_z/|\mathbf{A}|)\,\hat{k}$$
$$k\mathbf{A} \text{ has magnitude } |k|\,|\mathbf{A}|$$

Worked example — unit vector
- A = 3 î + 4 ĵ. Magnitude |A| = √(9 + 16) = 5.
- Unit vector Â = (3/5) î + (4/5) ĵ.

Final answer: Â = (3/5) î + (4/5) ĵ

4. Rectangular components and compass angles

In plain English: resolve means break a vector into axis pieces. θ is usually measured from +x. Watch the quadrant for signs. Compass wording: the last word is the reference line you start from.

What the words mean
- A_x = piece along x
- A_y = piece along y
- “30° east of north” → start from north, turn 30° toward east
- If θ is measured from +y, cos goes with y and sin with x

$$A_x = A\cos\theta$$
$$A_y = A\sin\theta$$
$$\mathbf{A} = A_x\,\hat{\imath} + A_y\,\hat{\jmath}$$
$$A = \sqrt{(A_x^2 + A_y^2)}$$
$$\tan\theta = A_y / A_x \text{ (choose the correct quadrant)}$$

Handy physics angles

sin 37° ≈ 3/5 , cos 37° ≈ 4/5

sin 53° ≈ 4/5 , cos 53° ≈ 3/5

sin 30° = 1/2 , cos 30° = √3/2

sin 45° = cos 45° = 1/√2

Sign by quadrant

1st: $A_x > 0$, $A_y > 0$

2nd: $A_x < 0$, $A_y > 0$

3rd: $A_x < 0$, $A_y < 0$

4th: $A_x > 0$, $A_y < 0$

Worked example — resolve a force
• F = 50 N at 37° above +x.
• $F_x = 50 \times (4/5) = 40$ N , $F_y = 50 \times (3/5) = 30$ N.
Final answer: F = 40 î + 30 ĵ (newton)

5. Triangle law and the resultant

In plain English: put the tail of B at the tip of A. The arrow from the free start of A to the free end of B is R = A + B. Drawing order does not change R. A + B = B + A.

What the words mean
• θ = angle between A and B
• α = angle that R makes with A
• Maximum R when same direction; minimum when opposite

$$\mathbf{R = \sqrt{(A^2 + B^2 + 2\,A\,B\cos\theta)}}$$
$$\mathbf{\tan\alpha = (B\sin\theta) / (A + B\cos\theta)}$$
$$\mathbf{|A - B| \le R \le A + B}$$
$$\mathbf{|A + B| \le |A| + |B|\ \text{(never bigger than the sum of sizes)}}$$

Special cases

θ = 0°: R = A + B (maximum)

θ = 180°: R = |A − B| (minimum)

θ = 90°: $R = \sqrt{(A^2 + B^2)}$

If |A| = |B| = A: $R = 2A\cos(\theta/2)$

If |A| = |B| = A: $|A - B| = 2A\sin(\theta/2)$

Equal vectors tip

When |A| = |B|, R bisects the angle

Three forces close a triangle if each

size lies between |diff| and sum of others

Polygon law: many vectors head-to-tail;

resultant joins first start to last tip

Remember: Cos θ is negative when θ is between 90° and 180°. That makes R smaller — which matches the picture.

6. Subtraction, component add, Δv

In plain English: subtraction is A + (−B). Flip B, then add. The safest everyday method: add x with x, y with y, z with z.

$$\mathbf{A - B = A + (-B)}$$
$$\mathbf{|A - B| = \sqrt{(A^2 + B^2 - 2\,A\,B\cos\theta)}}$$
$$\mathbf{R_x = A_x + B_x}$$
$$\mathbf{R_y = A_y + B_y}$$
$$\mathbf{R_z = A_z + B_z}$$
$$\mathbf{R = R_x\,\hat{i} + R_y\,\hat{j} + R_z\,\hat{k}}$$
$$\mathbf{\Delta v = v_f - v_i}$$

Remember: Rotation example: 90° anti-clockwise in xy-plane sends î → ĵ and ĵ → −î.

Worked example — add by columns
• $A = 2\hat{i} + 3\hat{j}$, $B = -\hat{i} + 4\hat{j}$.
• $R_x = 2 - 1 = 1$, $R_y = 3 + 4 = 7$.
• $R = \hat{i} + 7\hat{j}$, $|R| = \sqrt{(1 + 49)} = \sqrt{50} = 5\sqrt{2}$.
Final answer: $R = \hat{i} + 7\hat{j}$

7. Dot product, projection, angle tricks

In plain English: the dot product mixes two vectors and gives a number. It is large when they point the same way, zero when perpendicular, and negative when the angle is obtuse.

What the words mean
• $A \cdot B$ is a scalar (a number, with units of product of the two)
• Projection of B along A = $(A \cdot B) / |A|$
• $A \cdot B = B \cdot A$ (order does not matter)

$$\mathbf{A \cdot B = |A|\,|B| \cos\theta}$$
$$\mathbf{A \cdot B = A_xB_x + A_yB_y + A_zB_z}$$
$$\mathbf{\cos\theta = (A \cdot B) / (|A|\,|B|)}$$
$$\mathbf{A \cdot A = |A|^2}$$
$$\mathbf{\text{Projection of B on A} = (A \cdot B) / |A|}$$

Situation	What you get
θ = 0° (same way)	$A \cdot B = \|A\|\,\|B\|$ (maximum)
θ acute	$A \cdot B > 0$
θ = 90°	$A \cdot B = 0$
θ obtuse	$A \cdot B < 0$
θ = 180°	$A \cdot B = -\|A\|\,\|B\|$
Unit axes	$\hat{i} \cdot \hat{i} = 1$; $\hat{i} \cdot \hat{j} = 0$

Angle-condition shortcuts (equal sizes often)

Given	Angle between A and B
$\|A\| = \|B\| = \|A + B\|$	120°
$\|A\| = \|B\| = \|A - B\|$	60°
$\|A + B\| = \|A - B\|$	90°
$\|A + B\| = \|A\| + \|B\|$	0°
$\|A - B\| = \|A\| + \|B\|$	180°

Worked example — angle between two vectors
• $A = \hat{i} + \hat{j}$, $B = \hat{i} - \hat{j}$.
• $A \cdot B = 1 - 1 = 0$, so θ = 90°.
Final answer: θ = 90°

8. Cross product and $\hat{i}\,\hat{j}\,\hat{k}$

In plain English: the cross product gives another vector. Size uses sin θ. Direction is perpendicular to both (right-hand rule). Flip the order and the arrow flips.

What the words mean
• $|A \times B|$ = area of the parallelogram made by A and B
• Maximum size when θ = 90°: $|A \times B| = |A|\,|B|$
• Parallel or anti-parallel → cross product is zero

$$|\mathbf{A} \times \mathbf{B}| = |\mathbf{A}|\,|\mathbf{B}| \sin\theta$$
$$\mathbf{A} \times \mathbf{B} = -(\mathbf{B} \times \mathbf{A})$$
$$\text{Unit vector along } \mathbf{A} \times \mathbf{B} = (\mathbf{A} \times \mathbf{B}) / |\mathbf{A} \times \mathbf{B}|$$
$$\text{In xy-plane: } \mathbf{A} \times \mathbf{B} = (A_x B_y - A_y B_x)\,\hat{k}$$

Full 3D form

$$\mathbf{A} \times \mathbf{B} = (A_y B_z - A_z B_y)\,\hat{i} + (A_z B_x - A_x B_z)\,\hat{j} + (A_x B_y - A_y B_x)\,\hat{k}$$

Unit-vector cycle (must know)

$\hat{i} \times \hat{j} = \hat{k}$

$\hat{j} \times \hat{k} = \hat{i}$

$\hat{k} \times \hat{i} = \hat{j}$

Flip order → flip sign: $\hat{j} \times \hat{i} = -\hat{k}$

Same vector: $\hat{i} \times \hat{i} = 0$

Dot vs cross — one glance

Dot → a number (scalar)

Cross → a vector

Dot uses cos θ

Cross uses sin θ

Parallel → cross is zero

Perpendicular → dot is zero

Remember: Right-hand rule: fingers from A toward B (smaller angle). Thumb shows A × B.

Worked example — simple cross product
• $A = 2\,\hat{i}$, $B = 3\,\hat{j}$.
• $A \times B = 2\,\hat{i} \times 3\,\hat{j} = 6\,(\hat{i} \times \hat{j}) = 6\,\hat{k}$.
Final answer: $6\,\hat{k}$

9. Physics uses — work, torque, more

In plain English: work uses the dot product. Torque uses the cross product. Same idea appears in magnetic force and angular momentum.

Work (dot product)

$W = F \cdot s = F\,s \cos\theta$

θ = angle between F and displacement

If F ⊥ s, work = 0

Unit: joule (J)

Torque (cross product)

$\tau = r \times F$

$|\tau| = r\,F \sin\theta$

θ = angle between r and F

Direction: perpendicular to the plane

$$\text{Magnetic force: } \mathbf{F} = q\,(\mathbf{v} \times \mathbf{B})$$
$$\text{Angular momentum: } \mathbf{L} = \mathbf{r} \times \mathbf{p}$$

10. Revision lock lines — say these aloud

Topic	Say this in one breath
Write a vector	$\mathbf{A} = |A|\,\hat{A}$; also $A_x\,\hat{i} + A_y\,\hat{j} + A_z\,\hat{k}$
Magnitude	$|A| = \sqrt{(A_x^2 + A_y^2 + A_z^2)}$
Unit / scale	$\hat{A} = A/|A|$; kA flips if k is negative
Resolve	$A_x = A\cos\theta$, $A_y = A\sin\theta$
Resultant	$R = \sqrt{(A^2 + B^2 + 2AB\cos\theta)}$; $|A-B| \le R \le A+B$
Equal sizes	$R = 2A\cos(\theta/2)$; $|A-B| = 2A\sin(\theta/2)$

Topic	Say this in one breath
Add safely	Add x with x, y with y, z with z
Dot	A·B = AB cos θ = sum of component products
Projection	(A·B)/|A| is the piece of B along A
Perpendicular	A·B = 0 (neither zero)
Cross	|A×B| = AB sin θ ; A×B = −B×A ; RHR
$\hat{i}\ \hat{j}\ \hat{k}$	$\hat{i}\times\hat{j}=\hat{k}$, $\hat{j}\times\hat{k}=\hat{i}$, $\hat{k}\times\hat{i}=\hat{j}$
Work / torque	W = F·s ; τ = r×F

Last tip
Draw a small arrow sketch before you calculate.
Ask: do I need cos (dot / components) or sin (cross / perpendicular piece)?
If you cannot say a formula in one spoken sentence, that page needs one more read.

Source: Excellence Vector lectures 01–05. For revision and class notes — not a question paper.